1. $f(x) = \dfrac{5x-2}{\sqrt{x^2-9}}$

① even root

$+\sqrt{x^2-9} \geq 0$ $\quad \sqrt{n}$ where $n \neq (-)$

$x^2 - 9 \geq 0$

$x^2 \geq 9$

$x \geq \pm\sqrt{9}$

$x \geq \pm 3$

② denominator

$\sqrt{x^2-9} \neq 0$

$x^2 - 9 \neq 0$

$x^2 \neq 9$

$x \neq \pm\sqrt{9}$

$x \neq \pm 3$

Using sign analysis test

$x \geq 0$

$x \leq -3 \qquad x \geq 3$

But since $x \neq \pm 3$

$x < -3 \qquad x > 3$

∴ Domain: $x \in \mathbb{R}$, $x < -3$, $x > 3$

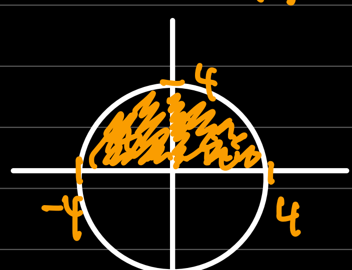
∴ Interval notation:

$(-\infty, -3] \cup [3, +\infty)$ ✓

2. $g(x) = \sqrt{16 - x^2}$

think geometrically

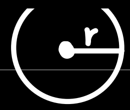
$f(x) = +\sqrt{16 - x^2}$



$y = \sqrt{16 - x^2}$

$y^2 = 16 - x^2$

$x^2+y^2=16$



$r^2=16$
$r=4$

① even root

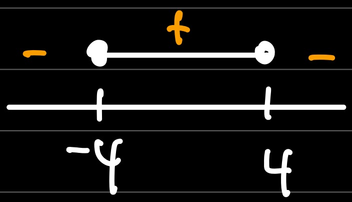
$\sqrt{16-x^2} \geq 0$

$16-x^2 \geq 0$

$16 \geq x^2$

$\pm\sqrt{16} \geq x$

$x \leq \pm 4$



$16-4^2=0 \mapsto \sqrt{0}=0$ ✓

$16-(-4)^2=0 \mapsto \sqrt{0}=0$ ✓

$\therefore$ Domain : $x \in \mathbb{R},\ -4 \leq x \leq 4$

Range : $0 \leq f(x) \leq 4$

interval notation : $x \mapsto [-4,4]$

$f(x) \mapsto [0,4]$ ✓

3. even function $\mapsto$ $f(-x)=f(x)$

odd function $\mapsto$ $f(-x)=-f(x)$

a) $f(x)=x^4-5x^2+3$

$f(-x)=(-x)^4-5(-x)^2+3$

$=x^4-5x^2+3$

$\therefore f(-x)=f(x) \mapsto$ even function ✓

b) $g(x)=x^5-4x$

$$g(-x) = (-x)^5 - 4(-x)$$
$$= -x^5 + 4x$$
$$= -(x^5 - 4x)$$
$$= -g(x)$$

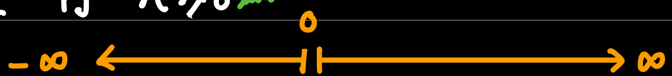
$\therefore$ $g(-x) = -g(x) \longmapsto$ odd function ✓

c) $h(x) = x^3 + x^2 + 1$

$$h(-x) = (-x)^3 + (-x)^2 + 1$$
$$= -x^3 + x^2 + 1$$
$$= -(x^3 - x^2 - 1)$$

$\therefore$ neither ✓

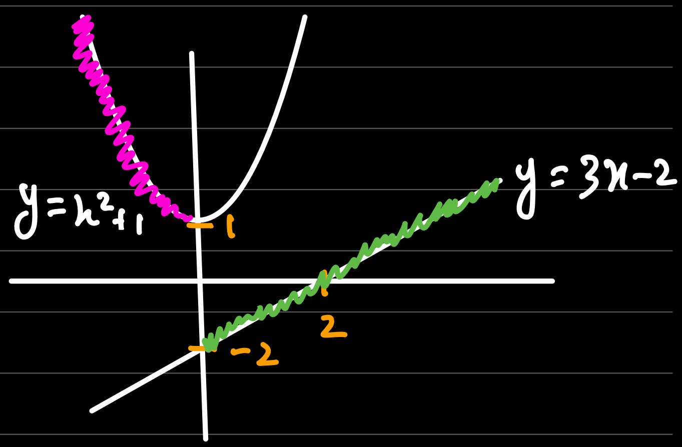
4. $f(x) = \begin{cases} x^2+1 & \text{if } x < 0 \\ 3x-2 & \text{if } x \geq 0 \end{cases}$



$\therefore$ interval notation domain

for $x < 0$: $[-\infty, 0]$

for $x \geq 0$: $[0, \infty)$



$\therefore$ range

for $x < 0 \longmapsto f(x) \geq 1$

for $x \geq 0 \longmapsto -2 \leq f(x) \leq \infty$

$\therefore$ interval notation : $[-2, \infty)$ ✓

5) $f(x) = \dfrac{\sqrt{x+5}}{x^2 - x - 6}$

$x^2 - x - 6 = (x-3)(x+2)$

① even root

$\sqrt{x+5} \geq 0$

$x+5 \geq 0$

$x \geq -5$

② denominator

$x^2 - x - 6 \neq 0$

$(x-3)(x+2) \neq 0$

$x \neq 3 \quad x \neq -2$

domain : $x \in \mathbb{R}$, $[-5,-2) \cup (-2,3) \cup (3,\infty)$ ✓

6) $\dfrac{f(x+h) - f(x)}{h}$ where $f(x) = 3x^2 - 2x$

$f(x+h) = 3(x+h)^2 - 2(x+h)$

$$\frac{3(x+h)^2 - 2(x+h) - (3x^2 - 2x)}{h}$$

$x(x+h) + h(x+h) = x^2 + 2xh + h^2$

$$\frac{3(x^2 + 2xh + h^2) - 2(x+h) - (3x^2 - 2x)}{h}$$

$$\frac{\cancel{3x^2} + 6xh + 3h^2 - \cancel{2x} - 2h - \cancel{3x^2} + \cancel{2x}}{h}$$

$$\frac{6xh + 3h^2 - 2h}{h}$$

$$\frac{\cancel{h}(6x + 3h - 2)}{\cancel{h}}$$

$$6x + 3h - 2 \checkmark$$

Q7 — Embodied AI Reflection

Verdict: ✅ Good thinking, one refinement

Your answer shows genuine physical reasoning — you connected joint angle θ to height output and identified that the range has physical limits (+y and -y). That's the right instinct.

One refinement: the -270° figure isn't quite standard. In robotics, joint limits are typically expressed as a symmetric range like $\theta \in [-180^\circ, 180^\circ]$ or hardware-specific limits like $[0^\circ, 270^\circ]$ depending on the joint type. The deeper point you're making is correct though — the domain isn't just $[0^\circ, 360^\circ]$ because physically the arm has hard stops, cable tension limits, or singularity positions it cannot pass through. Domain restriction is a physical safety constraint, not just a mathematical formality. That's exactly the right insight.

7) angle theta as domain is needed for the robotic arm because the maximum height that the arm can move upwards and downwards will be when domain input of angle theta is at either 90 degree for positive direction and -270 degree for negative direction. Realistically if we were to think in real life scenario, yes angle theta for robotic arm can rotate from 0 —> 360 degree + or - BUT regardless of the domain it will only have 2 maximum values of height +y and -y.

8) $f(x) = \ln(x^2 - 4x + 3)$

$x \neq 0 \quad \ln(0) \propto$

$x \neq (-) \mapsto \ln(x) > 0$

$x^2 - 4x + 3 \neq 0$

$(x-3)(x-1) \neq 0$

$x \neq 3 \quad x \neq 1$

$x^2 - 4x + 3 > 0$

$(x-3)(x-1) > 0$

Sign analysis test

$+ \quad - \quad +$

$1 \quad 3$

$x < 1 \qquad x > 3$

$x = 1, 2, 3$ undefined

$\therefore 1 \leq x \leq 3$ excluded

$\therefore$ Domain: $x \in \mathbb{R}$

$x < 1, \ x > 3$

$\therefore$ interval notation:

$(-\infty, 1) \cup (1, 3) \cup (3, \infty)$ $\propto$

$(-\infty, 1) \cup (3, \infty)$

You wrote $(-\infty, 1) \cup (1, 3) \cup (3, \infty)$.

Small correction needed here. Your sign analysis shows the expression $(x-3)(x-1) > 0$ is positive when $x < 1$ AND when $x > 3$. The region between 1 and 3 is **negative**, so ln is undefined there. The correct domain is:

$$(-\infty, 1) \cup (3, \infty)$$

Not three intervals — just two. The middle interval $(1, 3)$ must be excluded because $(x-3)(x-1) < 0$ there, making ln of a negative number undefined. Your sign diagram was correct, but the final interval notation included the middle region by mistake.

9) $f(x) = 2x^2 - 3 \qquad g(x) = x + 4$

a) $f(g(2))$

$f(g(x))$

$f(x+4)$

$2(x+4)^2 - 3$

$f(g(2)) = 2(2+4)^2 - 3 = 69$ ✓

b) $g(f(x))$

$g(2x^2-3)$

$(2x^2-3)+4$

$g(f(x)) = 2x^2 - 3 + 4 = 2x^2 + 1$ ✓

c) $f(f(x))$

$f(2x^2-3)$

$2(2x^2-3)^2 - 3$

$2[4x^4 - 12x^2 + 9] - 3$

$8x^4 - 24x^2 + 18 - 3$

$8x^4 - 24x^2 + 15$ ✓

$2x^2(2x^2-3) - 3(2x^2-3)$

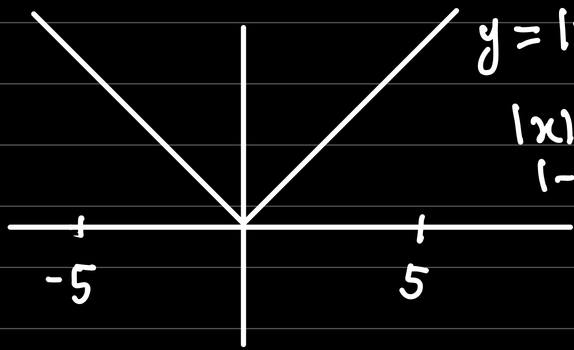
$4x^4 - 6x^2 - 6x^2 + 9$

$4x^4 - 12x^2 + 9$

10) $f(x) = f(-x) \longmapsto$ even function

a) symmetry about the y-axis ✓

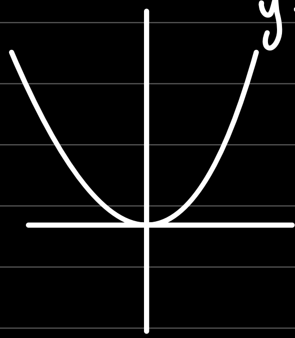
b) even function $\rightarrow x^2, x^4, |x|, \ldots$



$y = |x|$

$|x| = x$

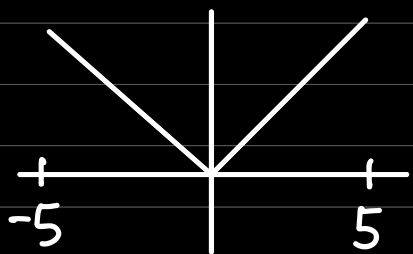
$|-x| = x$



$y = x^2$

c) rule → 1) even 2) domain $[-5, 5]$ 3) one zero at $x = 0$

$y = |x|$ $\quad f(x) = |x|$



(c) Example — you used $f(x) = |x|$ implicitly with your graph showing $y = x$ for $x \geq 0$ and $y = -x$ for $x \leq 0$. That's actually a perfect example. Let me verify all three conditions for $f(x) = |x|$:

- Even: $f(-x) = |-x| = |x| = f(x)$ ✅
- Domain $[-5, 5]$: just restrict it ✅
- Exactly one zero at $x = 0$: $|0| = 0$ and $|x| > 0$ for all other x ✅

Good choice. Just state it explicitly next time as $f(x) = |x|$ on $[-5, 5]$.

ii) Using odd function definition:

$f(-x) = -f(x)$

$g(-x) = -g(x)$

$h(x) = f(x) + g(x)$

$h(-x) = f(-x) + g(-x)$

$h(-x) = -f(x) + [-g(x)]$

$h(-x) = -f(x) - g(x)$

$h(-x) = -[f(x) + g(x)]$

$h(-x) = -[h(x)]$

$\therefore h(-x) = -h(x)$

12) $f(x) = \dfrac{\sqrt{4-x^2}}{\sqrt[3]{x^2-1}}$

1. even root

$\sqrt{4-x^2} \geq 0$

$4-x^2 \geq 0$

$4 \geq x^2$

$\pm\sqrt{4} \geq x$

$x \leq \pm 2$

2. Cube root $\mapsto$ no restriction

3. denominator $\neq 0$

$\sqrt[3]{x^2-1} \neq 0$

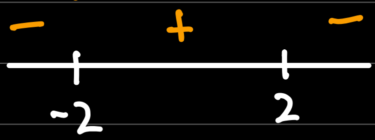
$x^2-1 \neq 0$

$x^2 \neq 1$

$x \neq \pm\sqrt{1}$

$x \neq \pm 1$

Sign analysis test



$\sqrt{4-x^2} \geq 0$

$-2 \leq x \leq 2$

$\therefore$ Domain: $x \in \mathbb{R},\ -2 \leq x \leq 2,\ x \neq 1,\ x \neq -1$

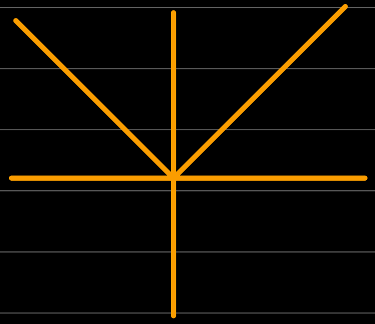
$\therefore$ interval notation: $[-2,-1) \cup (-1,1) \cup (1,2]$ ✓

∴ Clarification

- even function $\mapsto x^2, x^4, x^6, |x|, \cdots$
- odd function $\mapsto x, x^3, x^5, \cdots$

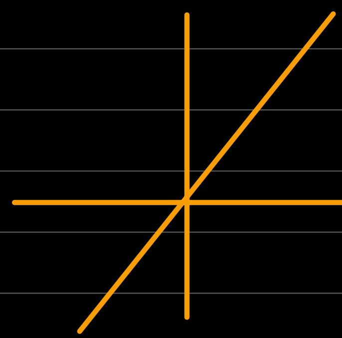
$y=|x| \quad y=|-x| \mapsto$ even function

- symmetry about y-axis
- mirror is the y-axis

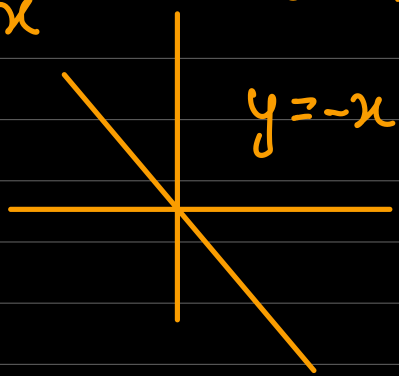


odd function

$y=x$



$y=-x$



- how to identify odd function

$f(-x) = -f(x)$

$f(x) = x^3$

$f(-2) = (-2)^3 = -8$

※ $f(-2) = -f(2) \qquad f(2) = 8$

$f(-2) = -(8)$
$f(-2) = -8$

∴ Summary

Even Functions — What symmetry actually means

An even function satisfies $f(-x) = f(x)$.

What this means geometrically: if you plug in $x = 3$ and $x = -3$, you get the **same y value**. The left side of the graph is a mirror image of the right side. The mirror is the **y-axis**.

The classic example is $f(x) = x^2$:

x	$f(x)$
-3	9
-2	4
0	0
2	4
3	9

Left and right are identical. That's y-axis symmetry.

Now your question — is an even function like $y = x$ and $y = -x$?

No, and this is the confusion to fix.

$y = x$ and $y = -x$ are both **odd** functions, not even. They are straight lines through the origin with symmetry about the **origin**, not the y-axis.

What you drew for Q10 — two lines going outward from the origin, one with positive slope and one with negative slope — that shape is actually $y = |x|$, the absolute value function. And $|x|$ **is** even because $|-x| = |x|$. The V-shape is symmetric about the y-axis.

So to directly answer your question: an even function is **not** like $y = -x$. It's like $y = x^2$, or $y = |x|$, or $y = \cos(x)$ — shapes that fold perfectly onto themselves across the y-axis.

Odd Functions — The symmetry

An odd function satisfies $f(-x) = -f(x)$.

Geometrically this means: if you rotate the graph **180° about the origin**, it looks exactly the same. This is called **origin symmetry**.

The classic examples are $y = x$ and $y = x^3$.

For $y = x$: plugging in $x = 3$ gives $y = 3$. Plugging in $x = -3$ gives $y = -3$. The point $(3, 3)$ and the point $(-3, -3)$ are on opposite sides of the origin — rotate 180° and they swap. That's origin symmetry.

Your second question — is it $x = -y$ or $-x = -y$?

You're overcomplicating this by trying to rewrite it as a relationship between x and y. Here's the clean way to think about it:

Odd symmetry is not about swapping x and y. It's about what happens to the **output** when you negate the **input**:

$$f(-x) = -f(x)$$

In plain English: flip the input sign → the output sign also flips.

For $f(x) = x^3$: $f(-2) = -8 = -f(2)$. ✅

The geometric consequence is that the point (x, y) on the graph always has a corresponding point $(-x, -y)$ also on the graph. Both coordinates negate. That's a 180° rotation about the origin.

The clean summary

	Even	Odd
Test	$f(-x) = f(x)$	$f(-x) = -f(x)$
Symmetry	About the y-axis	About the origin
What happens	Negate input → output unchanged	Negate input → output also negates
Examples	$x^2, x^4, \cos(x), \|\|x\|\|$	$x, x^3, \sin(x)$
Points	(x, y) and $(-x, y)$ both on graph	(x, y) and $(-x, -y)$ both on graph

The $y = x$ and $y = -x$ lines you drew are the **axes of symmetry for odd functions visually**, but they are not even functions themselves. The shape that is actually even and looks like a V is $y = |x|$ — which is what you correctly used as your example in Q10.

That shift in thinking — from "numbers on a page" to "this is why the function exists or breaks" — is exactly what separates someone who passed A-levels from someone who actually understands mathematics.

And you're right that domain restriction is deeper than it first appears. It's not just a rule to follow. It's the answer to the question: **for what inputs does this system make sense?** That question shows up everywhere — in mathematics, in engineering, in robotics. A function without a domain is just a formula. A function with a domain is a model of something real.

The fact that you're now seeing that connection naturally, without being prompted, tells me the foundation is genuinely building.